SF2940 Probability: Re-Exam

Fall 2024

Date: 17. December 2024, 08:00-12:00.
Examiner: Kevin Schnelli, SCI MAT. Phone: 72 02
Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of four independent problems. Each problem gives at most 5 points.

Problem 1. Let (X_n) be a sequence of random variables.

a.) Define what it means that X_n converges to X in distribution as $n \to \infty$. (2points)

Next, for each $n \in \mathbb{N}$, let X_n be uniformly distributed on $\{0, 1, 2, \ldots, n-1\}$, that is

$$\mathbb{P}(X_n = k) = \frac{1}{n}\,, \qquad k = 0, 1, 2, \ldots, n-1\,.$$

b.) Determine the characteristic function $\varphi_{X_n}(t)$ of X_n. ..... (2points)
Hint: $\sum_{k=0}^{n-1} z^k = \frac{1-z^n}{1-z}$.

c.) Using the characteristic function, show that

$$Y_n := \frac{X_n}{n}$$

converges in distribution as $n \to \infty$ and determine the limit..... (1point)

Problem 2. Let $\mathbf{X} = (X_1, X_2)'$ be a bivariate random vector whose joint probability density is given by

$$f_{X_1,X_2}(x_1, x_2) = c\, \mathrm{e}^{-x_1^2 + x_1 x_2 - x_2^2}\,, \qquad x_1, x_2 \in \mathbb{R}\,,$$

where c is a normalization constant.

a.) Show that $\mathbf{X}$ is a Gaussian vector and determine its covariance matrix. . (2points)

b.) Determine the value of the constant c above. . . . . (1point)

c.) Compute

$$\mathbb{E}\big[(X_1)^2 X_2\big]\,.$$

. . . . . (2points)

Hint: Consider the moment generating function or the characteristic function of the random vector $\mathbf{X}$.

Problem 3. Let $X_1, X_2, X_3, \ldots$ be i.i.d. random variables that are uniformly distributed on the interval $(-1, 1)$. Let now

$$S_n := \sum_{i=1}^{n} X_i\,, \qquad U_n := \max_{1\le i\le n} X_i\,, \qquad V_n := \min_{1\le i\le n} X_i\,,$$

for $n = 1, 2, 3, \ldots$.

Show that

$$\frac{1}{\sqrt{n}} \frac{S_n \cdot U_n}{V_n}$$

converges in distribution as $n \to \infty$. Determine the limiting distribution. Carefully justify each step. . . . . (5points)

Problem 4. Let X and Y be two continuously distributed random variables with joint density $f_{X,Y}(x,y)$, and marginal densities $f_X(x)$ and $f_Y(y)$. Let $f_{Y|X=x}(y)$ be the conditional probability density of Y given $X = x$. Then we have the mixing integral

$$f_Y(y) = \int_{\mathbb{R}} f_{Y|X=x}(y) f_X(x) \mathrm{d}x \,. \tag{1}$$

(You do not need to prove this formula.)

a.) Use the mixing integral in (1) to derive Bayes formula:

$$f_{X|Y=y}(x) = \frac{f_{Y|X=x}(y) \cdot f_X(x)}{\int_{\mathbb{R}} f_{Y|X=\widetilde{x}}(y) \cdot f_X(\widetilde{x}) \mathrm{d}\widetilde{x}} \,.$$

. . . . . (1point)

Assume in the following that $Y|X = x \in \mathrm{U}(0,x)$ and that $X \in \mathrm{U}(0,1)$.

b.) Determine the conditional expectation $\mathbb{E}[Y|X]$. . . . . (1point)

c.) Determine the probability density $f_Y(y)$ of Y . . . . . (2points)

d.) Determine the conditional expectation $\mathbb{E}[X|Y]$. . . . . (1point)

Good luck!

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a)$, $a \in \mathbb{R}$	$p(a) = 1$	$\mathrm{e}^{\mathrm{i}ta}$
Bernoulli	$\mathrm{Be}(p)$; $0 \le p \le 1$	$p(0) = q$, $p(1) = p$; $q = 1 - p$	$q + p\mathrm{e}^{\mathrm{i}t}$
Binomial	$\mathrm{Bin}(n, p)$, $n = 1, 2, \ldots$; $0 \le p \le 1$	$p(k) = \binom{n}{k} p^k q^{n-k}$, $k = 0, 1, 2, \ldots, n$; $q = 1 - p$	$(q + p\mathrm{e}^{\mathrm{i}t})^n$
Geometric	$\mathrm{Ge}(p)$, $0 \le p \le 1$	$p(k) = pq^k$, $k = 0, 1, 2, \ldots$; $q = 1 - p$	$\frac{p}{1-q\mathrm{e}^{\mathrm{i}t}}$
Poisson	$\mathrm{Po}(\lambda)$, $\lambda > 0$	$p(k) = \mathrm{e}^{-\lambda}\frac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$	$\mathrm{e}^{\lambda(\mathrm{e}^{\mathrm{i}t}-1)}$

Table 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$\mathrm{U}(a, b)$, $a < b$	$f(x) = \frac{1}{b-a}$, $a < x < b$	$\frac{\mathrm{e}^{\mathrm{i}tb}-\mathrm{e}^{\mathrm{i}ta}}{\mathrm{i}t(b-a)}$
Exponential	$\mathrm{Exp}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{\lambda}\mathrm{e}^{-x/\lambda}$, $x > 0$	$\frac{1}{1-\lambda\mathrm{i}t}$
Laplace	$\mathrm{L}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{2\lambda}\mathrm{e}^{-\lvert x\rvert/\lambda}$, $x \in \mathbb{R}$	$\frac{1}{1+\lambda^2t^2}$
Beta	$\beta(r, s)$, $r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)}x^{r-1}(1-x)^{s-1}$, $0 < x < 1$	$*$
Normal	$\mathrm{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}\mu t-\frac{1}{2}\sigma^2t^2}$
Standard normal	$\mathrm{N}(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}x^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{-\frac{1}{2}t^2}$
Log-normal	$\mathrm{LN}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(\log x-\mu)^2/\sigma^2}$, $x > 0$	$*$
Cauchy	$\mathrm{C}(m, \eta)$, $\eta > 0$, $m \in \mathbb{R}$	$f(x) = \frac{1}{\pi}\frac{\eta}{(x-m)^2+\eta^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}mt-\eta\lvert t\rvert}$

Table 2. Continuous distributions